

ELBOs — six batch-conditioning arms

i indexes cells, g indexes genes, b_i is the batch label, θ_g the per-gene negative-binomial dispersion, β the v-term weight. Expectations are over the guide’s reparameterised draws.

model1-add — Decipher-VZ-Add

$q(z | x) q(v | z, x)$; batch on the z prior

$$\text{ELBO}_1 = \sum_i \mathbb{E}_{q(v_i, z_i)} \left[\sum_g \log p(x_{i,g} | z_i, \theta_g) + \log \frac{p(z_i | v_i, b_i)}{q(z_i | x_i)} + \beta \log \frac{p(v_i)}{q(v_i | z_i, x_i)} \right]$$

model2-add — Decipher-VZ2-Add

$q(z | x, b) q(v | z, x)$; batch on both z edges

$$\text{ELBO}_2 = \sum_i \mathbb{E}_{q(v_i, z_i)} \left[\sum_g \log p(x_{i,g} | z_i, \theta_g) + \log \frac{p(z_i | v_i, b_i)}{q(z_i | x_i, b_i)} + \beta \log \frac{p(v_i)}{q(v_i | z_i, x_i)} \right]$$

model3-add — Decipher-MF-Add

mean-field v : $q(v | x)$

$$\text{ELBO}_3 = \sum_i \mathbb{E}_{q(v_i, z_i)} \left[\sum_g \log p(x_{i,g} | z_i, \theta_g) + \log \frac{p(z_i | v_i, b_i)}{q(z_i | x_i, b_i)} + \beta \log \frac{p(v_i)}{q(v_i | x_i)} \right]$$

The three new arms move b_i out of the z terms and into the likelihood. Every b_i that moved is shown in **red**.

model4 — Decipher-ZX

$q(z | x) q(v | z, x)$; batch on the decoder

$$\text{ELBO}_4 = \sum_i \mathbb{E}_{q(v_i, z_i)} \left[\sum_g \log p(x_{i,g} | z_i, \textcolor{red}{b}_i, \theta_g) + \log \frac{p(z_i | v_i)}{q(z_i | x_i)} + \beta \log \frac{p(v_i)}{q(v_i | z_i, x_i)} \right]$$

model5 — Decipher-ZX2

$q(z | x, b) q(v | z, x)$; batch on the decoder

$$\text{ELBO}_5 = \sum_i \mathbb{E}_{q(v_i, z_i)} \left[\sum_g \log p(x_{i,g} | z_i, \textcolor{red}{b}_i, \theta_g) + \log \frac{p(z_i | v_i)}{q(z_i | x_i, b_i)} + \beta \log \frac{p(v_i)}{q(v_i | z_i, x_i)} \right]$$

model6 — Decipher-MF2

mean-field v ; batch on the decoder

$$\text{ELBO}_6 = \sum_i \mathbb{E}_{q(v_i, z_i)} \left[\sum_g \log p(x_{i,g} | z_i, \textcolor{red}{b}_i, \theta_g) + \log \frac{p(z_i | v_i)}{q(z_i | x_i, b_i)} + \beta \log \frac{p(v_i)}{q(v_i | x_i)} \right]$$

Reading the table. Exactly three slots vary: whether b_i appears in the likelihood $p(x_{i,g} | \cdot)$, in the z prior $p(z_i | \cdot)$, and in the z guide $q(z_i | \cdot)$. A fourth slot—whether the v guide reads z_i —separates the mean-field arms (3 and 6) from the rest.

Why arms 4–6 behave differently. In arms 1–3 the batch parameter appears *only* inside $p(z_i | v_i, b_i)$. The z_i reaching the likelihood is the guide’s draw, which contains no batch parameter, so $\partial \log p(x_{i,g} | z_i, \theta_g) / \partial \text{batch_shift} = 0$ exactly, on every sample: the reconstruction term cannot see it. In arms 4–6 the batch parameters sit inside $p(x_{i,g} | z_i, \textcolor{red}{b}_i, \theta_g)$ —the largest term in the objective.